

Problem statement

Retail and e-commerce businesses face difficulties in managing their supply chains because **demand is uncertain, inventory levels fluctuate, suppliers may be unreliable, and transportation can be inefficient.** These challenges can lead to **stockouts, excess inventory, delivery delays, and increased operational costs.**

The proposed project aims to develop an **AI-powered Smart Supply Chain and Inventory Optimization System** that can **predict future demand, identify supply-chain risks, optimize inventory and replenishment decisions, evaluate suppliers, and optimize logistics.** The system will use AI/ML and an agentic decision-making layer to provide **intelligent, actionable recommendations** that help businesses make better supply-chain decisions.

Businesses cannot accurately predict future product demand, which can cause **stockouts or excess inventory.**

Poor inventory planning and unreliable suppliers can lead to **shortages, supplier delays, quality issues, and increased costs.**

Inefficient route and vehicle planning can result in **higher transportation costs and delivery delays.**

SOLUTION

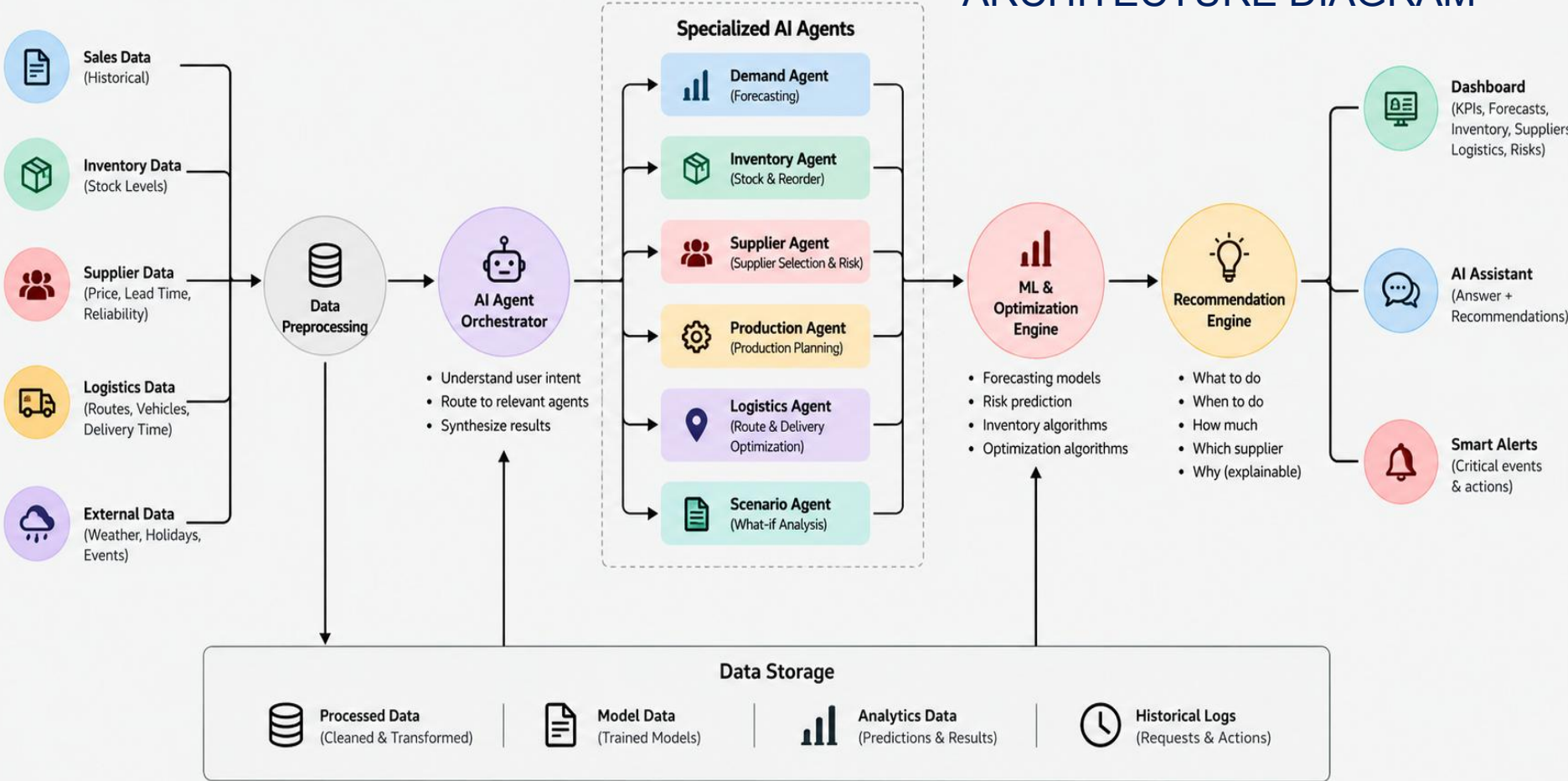
Use historical sales, seasonality, promotions, holidays, price, and location data to predict future demand and reduce uncertainty.

Analyze stock levels and calculate **safety stock, reorder points, and EOQ** to determine **when and how much to replenish**, reducing stockouts and excess inventory.

An **AI Agent Orchestrator** coordinates specialized agents and ML/optimization models to convert predictions into actionable recommendations, alerts, and what-if analysis for business users.

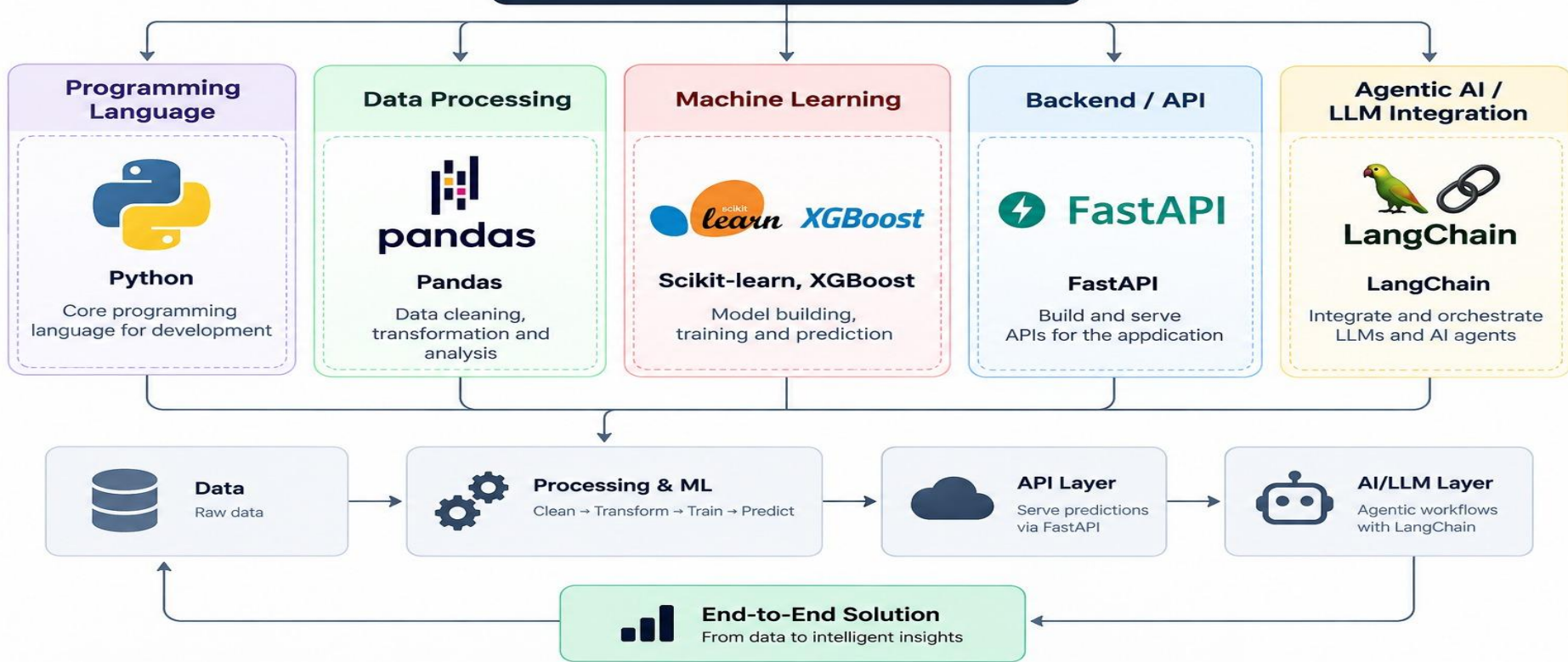
Evaluate suppliers based on **price, lead time, reliability, quality, and capacity**, while optimizing routes, vehicles, delivery time, and transportation costs.

ARCHITECTURE DIAGRAM



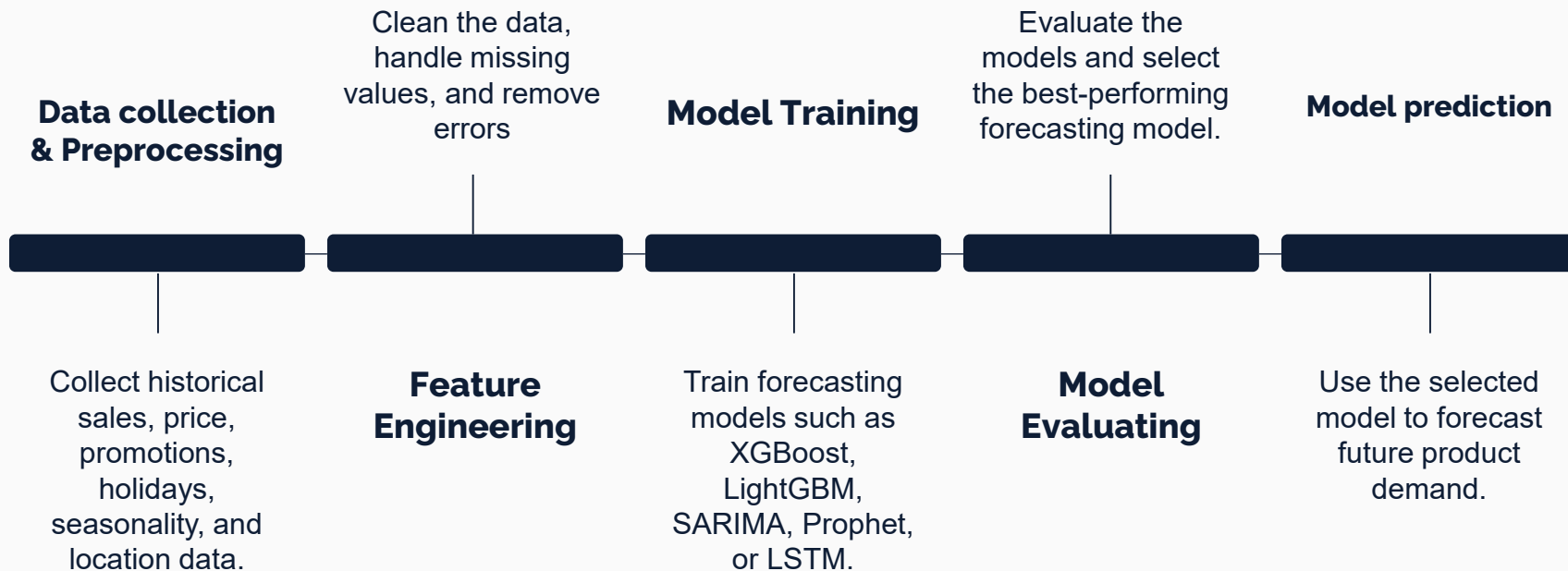


Tech Stack



DataSet: https://www.census.gov/retail/marts/historic_releases.html

Demand Prediction Model Process



Literature Review						
No.	Research Paper	Year	Author s	Journal/ Conferen ce	Problem Addressed	Method/Alg orithm Used
1	A simulation-based hybrid causal predictive framework for stockout risk analysis in supply chain	2026	Hafaf et al	PLoS ONE	Stockout risk prediction and causal effect of lead time	PSM, IV-2SLS, IPW, DRE, Random Forest, LightGBM, Moving Average
2	Explainable artificial intelligence to improve the resilience of perishable supply chains	2024	Elluru et al	Annals of Operations Research (Springer)	Customer-centric demand forecasting for perishable products	K-means, hierarchical clustering, XAI (SHAP), SVR, LSTM, VAR
3	Predictive models for inventory optimization: a machine learning application for demand forecasting at a Brazilian SME	2026	Multiple authors	Future Business Journal (Springer)	Demand forecasting for inventory optimization in SME	Amazon Forecast (DeepAR+), SageMaker Canvas AutoML

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4	Deep Policy Iteration with Integer Programming for Inventory Management	2025	Bhat et al.	Manufacturing & Service Operations Management (INFORMS)	Network inventory replenishment with combinatorial action space	Programmable Actor RL (PARL), deep policy iteration + integer programming
5	A Deep Q-Network for the Beer Game: Deep Reinforcement Learning for Inventory Optimization	2021	Oroojzadeh et al.	Manufacturing & Service Operations Management (INFORMS)	Multi-echelon inventory optimization (bullwhip effect)	Deep Q-Network (DQN)

Revenue Generation Model

Revenue Source	How Revenue is Generated
SaaS Subscription	Monthly or annual subscription based on company size, number of products, users, or transactions
Premium Plans	Additional charges for advanced forecasting, supplier-risk analysis, and autonomous decision-making
Enterprise Solutions	Customized pricing for large organizations requiring advanced features and higher usage
API / Integration Services	Charges for integrating the platform with ERP, e-commerce, inventory, and logistics systems
Implementation & Support	One-time implementation fees and ongoing technical/support services

THANK YOU